

# Responsible Implementation of AI Tools in Technical Documentation Workflows

*Russell A. Spangler*

## Introduction

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AI is a genuinely useful tool for technical documentation work, but it fails in specific and predictable ways, and I have come to believe that using it responsibly is less about trusting the tool and more about designing your workflow around how it breaks. I have used these tools extensively over the past two years, across drafting, editing, research, and analysis, and the clearest lesson I can offer is this: it requires, at least, a babysitter, someone who checks its work, catches what it misses, and knows enough about the underlying material to recognize when the output only looks right.

That last part matters more than people tend to assume. AI-generated content, whether a first draft or a compliance summary, can be polished and confident and still be wrong in ways not obvious on a first read, particularly when the person reviewing it does not already know the source material well. I have seen this often enough, in repeatable ways, that I no longer think of it as an occasional failure. I think of it as a structural feature of how these tools work, one that any disciplined implementation has to plan for rather than hope to avoid.

This paper is not an argument against using AI in technical documentation work. I use it every day, and I believe it is one of the more valuable tools available to a technical writer or editor right now, one that is only going to become more capable. But usefulness and trustworthiness are two different questions, and conflating them is, in my experience, the most common mistake teams make when adopting these tools. What follows is an account of where AI earns trust, where it does not, and what I think a prudent implementation actually requires, drawn from specific things I have watched it get right, and specific things I have watched it get wrong.

None of this means AI replaces the thinking. It cannot replace human judgment, creativity, or problem-solving, and I don't think that is a temporary limitation so much as a basic feature of what these tools are. What it does replace, or at least significantly reduce, is labor, particularly the repetitive kind: finding things, organizing things, restating or summarizing things, or checking things against a standard. That distinction, between reducing labor and replacing thought, is where I think most AI implementation programs fail. Not because the tools don't work, but because the people implementing them misunderstand what they actually are.

## Where AI Earns Trust: Extraction, Analysis, and Review

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The clearest pattern I have found, across every tool I have used, is that AI is at its best when the job is to find and organize something that already exists, and at its most unreliable when the job is to invent something that does not. Extraction, analysis, summarization, first pass review of already-written material: this is where these tools consistently earn their keep, and where I trust them the most.

A chat-based interface is the most flexible version of this. It is session-based, so I can revisit prior work without re-explaining the whole context, and it rewards a clear prompt paired with current

source material about as reliably as anything I have used. Give it a well-defined document and a specific question, and it is fast, and genuinely useful for early-stage thinking. Ask it to work from memory or a vague prompt instead of an actual source document, and the quality drops off fast. That's really just a version of the same lesson: the tool is only as good as what you feed it.

Prompt libraries, pre-configured assistants, and deep-research modes are all, in one way or another, attempts to make that first pattern more repeatable, and they help, up to a point. What actually drives good output, in my experience, is not the library or the assistant itself but the discipline underneath it: a prompt that clearly states the context, the task, and the expected output, rather than a vague ask and a hope. That structure is worth building into a habit regardless of which tool delivers it. I'm more skeptical of pre-built assistants that hard-code the steps of that structure into a fixed sequence you cannot deviate from, even when the situation calls for it. A good assistant, built for one specific job, such as identifying and organizing a list of requirements from a solicitation, tends to be more consistent than asking the same question freeform. But rigidity in the tool, as opposed to rigidity in how carefully you structure your ask, cuts both ways: it walks you through the same set of questions every time, which is great for consistency and repeatability. But when you just wanted a quick answer, a skilled prompter working directly in chat, applying that same context-task-output discipline by hand, can often match or beat the tool in both speed and flexibility.

Review tools follow the same logic. Asked to tighten language, catch inconsistency across multiple authors, or flag places where wording drifts from a defined standard, AI review is efficient and genuinely useful, and it is one of the few places I would call it close to reliable. But its usefulness is bounded entirely by what standard it is given. Without clear evaluation criteria or instructions, it drifts toward commenting on style rather than substance, and it should never be treated as a stand-in for an actual compliance check. It edits well. It does not certify.

What ties all of this together, for me, is that none of it asks the model to create anything new. It is finding, organizing, comparing, and tightening material that already exists, a fundamentally lower-risk task than generating original content, which is where things start to go wrong.

## Where It Breaks: Generation and Unverified Output

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None of this is an argument that AI is bad at generating a first draft. It is often quite good at one, and I use it regularly for exactly that: getting a rough version of something on the page fast, or working through a brainstorm when I am not yet sure what shape a document should take. That kind of early-stage generation is genuinely useful, and I would not want to talk anyone out of it. That said, the results are more useful when the model has a wealth of context documents to draw information and style from.

The problem starts when a generated draft is mistaken for a finished one. A first draft, AI-written or otherwise, is not something you copy, paste, and submit. The risk with AI-generated drafts specifically is that they tend to look more finished than they are and can hide a number of issues. The prose is clean, the structure is confident, the tone is consistent, and all of that polish can make a reader, including the person who requested the draft, forget that none of it has been checked against anything yet.

That risk grows the further a task moves away from working with real source material toward inventing content outright. I once watched a drafting tool produce a proposal narrative that read as competent and specific, right up until I checked it against what information we actually had on hand. Entire past-performance projects and personnel had been invented to cover gaps the tool didn't have real information to fill. Nothing in the writing gave it away. The fabricated sections read exactly like the true ones.

This is, I think, the sharpest version of the distinction between reducing labor and replacing thought. A tool that gets you to a rough draft faster is reducing labor, and that is valuable. A tool that quietly manufactures facts to make a draft look complete is not doing your thinking for you; it is doing something closer to guessing, confidently, and leaving you to find out later which parts were real. The more a task depends on the model inventing content rather than working from something already in front of it, the more that risk grows, and the more scrutiny the output needs before anyone treats it as usable.

### Named Failure Patterns

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The clearest way I can make the case for validation is to show, specifically, where these tools have gotten it wrong for me. Not hypothetically, but in actual solicitation documents, on actual proposals, in ways that would have gone out the door wrong if nobody had checked.

The first pattern is flattening conditional language into something simpler and, in the process, wrong. I recall a solicitation that required the program manager to hold a specific degree and certification, or, in lieu of one or both, a certain number of years of experience. When I asked an AI tool to extract the requirement, it reported back that the degree and certification were required, full stop. The substitution clause, the “or” that made the requirement negotiable, simply disappeared. Read on its own, the AI's summary sounded confident and complete. It was wrong in a way that could have disqualified a perfectly qualified candidate from consideration, had anyone acted on it without checking the original text.

The second pattern is collapsing two related but distinct numbers into one. A different solicitation required project descriptions completed within the last five years, and, separately, references from within the last three years: two different windows for two different requirements. The AI's summary reported a single number, three years, applied to both. Nothing about the summary looked incomplete. It read like a clean, simple requirement, which was exactly the problem, since the actual requirement was neither clean nor simple.

What connects these two examples, for me, is that neither failure looked like a failure. There was no error message, no hedge, no visible seam where the model was unsure. Both outputs read as confident, finished, and correct, and the only way to catch either mistake was to already know the source document well enough to notice that something had gone missing. That is, I think, the real argument for validation: not that AI is unreliable in some abstract sense, but that its errors are specifically the kind that look right until someone checks.

## What a Sound Implementation Actually Requires

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Everything above points toward the same practical conclusion: safe use of these tools is not a matter of trusting them more or less, but of building a workflow that accounts for exactly how and where they fail. In my experience, that comes down to a handful of concrete requirements, not a general attitude of caution.

A sound implementation requires:

- Treating any AI-generated content touching compliance, requirements, or contractual language as a draft in need of verification, never as a finished output, regardless of how polished it looks;
- Feeding the tool complete, current source material rather than relying on it to work from memory or a vague prompt, since output quality is bounded almost entirely by input quality;
- Drawing a clear line between tasks that ask the model to find, organize, or tighten existing material, which it does well, and tasks that ask it to invent new content, which carry meaningfully higher risk and need a correspondingly higher level of review;
- Building prompts around a consistent structure, context, task, and output, rather than depending on a pre-built assistant or tool to enforce that discipline for you; and
- Assigning specific human ownership for verification, so that checking the work is a defined responsibility and not something everyone assumes someone else is doing.

None of this is complicated, and none of it requires slowing down. What it requires is treating the tool as exactly what it is: something that reduces labor, substantially, and does not replace the judgment of the person still responsible for the work.

## Conclusion

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None of this is a case against using AI in technical documentation work. I use these tools every day, and they have made real parts of my job faster and more consistent, particularly the repetitive, first-pass work that used to eat the most time before I could get to the parts that actually needed my judgment. The tools are good, and they are getting better quickly.

But good and finished are not the same thing, and neither are fast and safe. Every failure I have described here, the flattened exception, the collapsed date range, the invented past performance, looked complete right up until someone who knew the material checked it. That is the part a lot of implementation efforts still miss: the danger is not that AI produces obviously bad work. It is that it produces work that looks exactly as good as work that has already been checked, whether or not it actually has been.

A tool that reduces labor is worth adopting. A tool mistaken for a substitute for judgment is worth worrying about. The case I am making here is that those are two different tools, built on the same technology, and the difference between them comes down entirely to how carefully the people using it choose to treat it.